Linear Search in C - Training Attendance

```
#include <stdio.h>

#define M 100

int linearSearch(int arr[], int n, int key)
{
    int i;

    for (i = 0; i < n; i++)
    {
        if (arr[i] == key)
        {
            return i;
        }
    }

    return -1;
}

int main()
{
    int roll[M];
    int n, i, choice, key, result;

    printf("Enter the number of students attended training: ");
    scanf("%d", &n);

    if (n < 1 || n > M)
    {
        printf("Invalid number of students!\n");
        return 1;
    }

    printf("Enter roll numbers:\n");

    for (i = 0; i < n; i++)
    {
        scanf("%d", &roll[i]);
    }

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Display Roll Numbers\n");
        printf("2. Search Roll Number\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("\nStudents Attended Training:\n");

                for (i = 0; i < n; i++)
                {
                    printf("%d ", roll[i]);
                }

                printf("\n");
                break;
```

```
            case 2:
                printf("Enter roll number to search: ");
                scanf("%d", &key);

                result = linearSearch(roll, n, key);

                if (result != -1)
                {
                    printf("Student with Roll Number %d attended "
                           "the training program.\n", key);
                }
                else
                {
                    printf("Student with Roll Number %d did NOT "
                           "attend the training program.\n", key);
                }

                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}
```